

Timing Search[†]

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Abstract

Why do industries fluctuate at different frequencies? I study a timing search model in which entrants choose dates and rapid industry reconfiguration strains a shared pipeline. Equal values across active entry dates hold effective profit at the entry cost, which ties the speed of reconfiguration to the industry's scale. The model admits multiple equilibria: the industry may rest at either break-even scale, cycle between them, or wither away. I focus on the cyclical equilibrium, the unique dynamically robust full-support equilibrium, which is also relevant to the observed dispersion in frequencies. Every interior state rotates at $\omega = \sqrt{\varphi/\mu}$, linking cycle frequency to the industry-level profit curvature φ and the timing friction μ . Finally, I derive an analytical state-dependent Pigouvian entry wedge and study its policy implications.

Keywords: Timing search, equilibrium multiplicity, industry frequency, entry timing, Pigouvian taxation.

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1 Introduction

Why do industries fluctuate at different frequencies? U.S. production and establishment entry exhibit substantial cross-industry dispersion in their dominant cycle frequency despite common aggregate conditions. Standard business-cycle frameworks usually tie cycle dynamics to a common shock process (Kydlan and Prescott, 1982) or a common policy rule. This paper asks whether industry-specific market structure helps set that frequency.

The mechanism is *timing search*. Entrants choose dates. A date with many entrants becomes crowded because firms draw on the same entry pipeline: permits, specialist labor, equipment, and financing. Time matters because an entrant can wait, but cannot move backward: each date inherits the industry created by earlier entry. In equilibrium, no entry date can offer a strictly better payoff than another; if one did, entrants would move toward it until the advantage disappeared. This date-by-date indifference is the temporal counterpart of the equal-value condition in search (Moen, 1997; Burdett and Mortensen, 1998; Wright et al., 2021), and an intertemporal-arbitrage condition whose closest formal precedent is Hotelling (1931). When the industry adjusts quickly, entrants strain the shared pipeline all at once, and per-firm profit is lower than when it moves slowly. What an entrant collects at date t therefore depends on the pace of adjustment $\dot{n}_t$, not on how many firms are already active. In equilibrium, indifference across different entry dates makes this pace depend on the industry’s scale: it moves fastest where profit is highest, at the optimal scale, and slows to a halt where profit falls to the entry cost. It cannot rest at the peak; it sweeps through and reverses at each break-even scale, leading to the endogenous cycle. Semiconductors illustrate it, sharing a slow entry pipeline that stretches a profitable rush into a glut and the correction into scarcity. The industry’s own profit landscape and entry bottleneck set how fast this repeats. The state is the mass of active establishments; per-establishment profit is single-peaked in it, thick at low scale and congested at high scale, and a symmetric friction makes effective profit fall with the speed of net adjustment.

The contributions are threefold. First, I characterize the endogenous cycle that arises from timing search and show that its frequency is set by industry-specific primitives (Theorem 1, Section 3). Under the log-quadratic benchmark the industry follows a harmonic path with closed-form angular frequency

$$\omega = \sqrt{\varphi/\mu}, \tag{1}$$

where φ is the curvature of profits around optimal scale and μ is the reduced-form timing friction. The cycle coexists with rest at a break-even scale and with withering, but it is the unique dynamically robust full-support equilibrium (Proposition 2) and the one relevant to the observed dispersion in frequencies.

Second, I take that branch’s two predictions to U.S. manufacturing (Section 4). Frequencies should differ across industries, and they should rise with profit curvature. Both run into the precision of the frequency estimates themselves: an industry’s own fitted dynamics reproduce its estimated peak only to within a factor of about 1.6, so the variation across industries is of the same order as the variation within them. The ordering across industries is consistent with the first prediction but does not separate them individually. Applying one spectral rule to all 21 NAICS-3 industries, with no screens, the slope of log frequency on concentration is positive and indistinguishable from zero; annual establishment frequencies share its sign, quarterly counts do not. The measurement precision and the concentration proxy both bound what these exercises can establish, and Sections 4.1–4.2 are explicit about it.

Third, private date choice equates an entrant’s own value across dates but ignores what scale and reconfiguration do to incumbents. I derive the state-dependent Pigouvian entry wedge that decentralizes an interior planner allocation (Proposition 3, Section 5). The simpler flat tax that collapses the private locus to $(n_p, 0)$ turns out to permit extinction from any inherited scale, so policy inherits the same multiplicity as the positive model.

Related literature This paper builds on four literatures.

Endogenous cycles. The plucking tradition interprets recessions as departures below a ceiling and emphasizes deeper-recession–faster-recovery patterns (Friedman, 1993; Bordo and Haubrich, 2017; Dupraz et al., 2019). That tradition takes the size of the departure as given and studies the return; free entry here pins the size itself, so an industry has a single orbit rather than a family indexed by the size of a disturbance. Beaudry et al. (2020, 2024) obtain continuing stochastic motion near a Hopf bifurcation. Timing search instead classifies a conservative equilibrium correspondence that also contains two endpoint rest paths.

An interaction-based tradition generates aggregate fluctuations by coordinating discrete adjustments (Nirei, 2006, 2015; Nirei and Scheinkman, 2024). Timing search instead uses congestion across neighboring dates: complementarity clusters adjustment into spikes, whereas congestion spreads entry over time. Shleifer (1986) also obtains cycles from adjustment timing, but through demand complementarity that synchronizes firms into one economy-wide cycle.

Models of animal spirits and self-fulfilling prophecies generate fluctuations from indeterminacy and beliefs (Farmer and Guo, 1994; Farmer, 1999); earlier cycle precedents include Grandmont (1985), Azariadis (1981), and Benhabib and Farmer (1994). Closest on timing, Jovanovic (2009) studies the option value of investment dates. The present paper instead studies equal values across all active dates and classifies the resulting paths.

Entry timing. The entry margin is strongly procyclical and propagates aggregate shocks (Lee and Mukoyama, 2015; Clementi and Palazzo, 2016). A complementary literature makes

when to enter a choice. The option to delay changes the opportunity cost of entry (Vardishvili, 2026); time to build propagates business cycles (Cook, 2001); and sunk entry with a lag transmits shocks through product variety (Bilbiie et al., 2012). Timing search shares the timing margin but locates its contribution in equal values across active dates and the associated equilibrium classification.

Search and adjustment frictions. Timing choice is in the spirit of Stigler (1961), McCall (1970), and especially the equal-value condition in Burdett and Mortensen (1998). Dates, however, are inherited states rather than market-clearing submarkets. The reduced-form pipeline friction connects to matching and adjustment-cost approaches (Diamond, 1982; Mortensen and Pissarides, 1994; Pissarides, 2000; Caballero and Hammour, 1994), but no matching market-clearing equation is imposed.

Industry dynamics. The model isolates timing within an endogenous-entry environment and is complementary to the heterogeneity and selection mechanisms of Hopenhayn (1992); Hopenhayn and Rogerson (1993). Startup cohorts inherit conditions at birth in Sedláček and Sterk (2017) and Pugsley et al. (2021); here each date inherits the stock created by earlier cohorts. Work on agglomeration motivates the single-peaked profit landscape (Krugman, 1991; Duranton and Puga, 2004; Rosenthal and Strange, 2004; Syverson, 2004).

2 Model

The environment has three elements: a hump-shaped profit landscape, a reduced-form timing friction, and a linear-utility household. A matching calculation provides an illustrative calibration of the friction but is not part of equilibrium market clearing.

2.1 Agents

The economy has two types of agents. A representative household consumes and supplies labor. A finite reservoir of potential entrepreneurs is either outside or operating a firm; while outside, an entrepreneur can wait before entering. The household has linear utility in consumption and linear disutility from labor, $U = \int_0^\infty e^{-\rho t} [C_t - \bar{w} L_t] dt$, so the static labor-supply condition pins $w = \bar{w}$ and labor is supplied elastically.¹ I impose the common deterministic discount rate $r = \rho > 0$, so private and planner discounting coincide. Three ingredients define the entry side of the environment.

¹This quasilinear specification is in the spirit of Greenwood et al. (1988) preferences in eliminating the wealth effect on labor supply.

Assumption 1 (Entry economy).

A mass $M > 0$ of potential entrepreneurs is either outside or active; an outside entrepreneur may wait costlessly and choose a later entry date. The outside option is zero, and participation into timing choice is competitive whenever the reservoir is not exhausted. Active firms produce using labor hired from the representative household at wage $\bar{w}$, with technology $y_i = z l_i^\gamma$, $\gamma \in (0, 1)$. Each entry uses $\kappa > 0$ units of real entry resources, paid at the moment of entry. Firms exit at Poisson rate $\delta > 0$, returning their entrepreneur to the outside pool.

Let n_t denote the mass of active production units, with $n_t < M$ on the full-support paths studied below. The outside mass supplies additional participants if entry value exceeds its real resource cost; interior participation therefore gives zero net surplus while the reservoir remains slack. The household ultimately owns the operating claims and bears the entry-resource outlay κ . Waiting has no flow payoff. I interpret n as active *establishments* rather than legal firms, matching the aggregate-entry interpretation below.

2.2 Scale efficiency and the profit landscape

Each firm's effective output depends on market thickness. More firms create denser networks of suppliers, customers, and complementary services, improving efficiency. But beyond a point, additional firms congest the market. A scale efficiency function $m(n)$ multiplies each firm's output: $y_i = z l_i^\gamma \cdot m(n)$, with productivity z and labor l_i . Because firms reoptimize labor against the scale effect, optimized profit raises it to the power $1/(1 - \gamma)$: $\pi(n) = \bar{\pi}(\bar{w}) m(n)^{1/(1-\gamma)}$, where $\bar{\pi}(\bar{w})$ is the standard per-firm profit at the fixed wage $\bar{w}$ of Section 2.1, earned absent any scale effect.²

The classification requires only a single-peaked reduced-form profit curve.

Assumption 2 (Single-peaked profits).

The profit function $\pi : [0, M) \rightarrow \mathbb{R}_{++}$ is continuous at zero and C^2 on $(0, M)$. It has a unique peak n_p : $\pi'(n) > 0$ for $n < n_p$ and $\pi'(n) < 0$ for $n > n_p$. Let $\bar{y} \equiv (r + \delta)\kappa < \pi(n_p)$, and suppose $\pi(n) = \bar{y}$ has two solutions $0 < n_- < n_p < n_+ < M$.

Single-peakedness allows productive thickness at low scale and congestion at high scale. It also makes the two break-even crossings transverse, a condition that ensures finite travel time at the turning points. The productivity-density evidence in Syverson (2004) motivates non-monotone scale effects, but does not identify the shape of this aggregate profit curve.

The log-quadratic case gives the sharpest benchmark:

$$\pi(n) = \pi_0 \exp\left(-\frac{\varphi}{2}(n - n_p)^2\right), \quad \pi_0 > 0. \quad (2)$$

²The static labor choice gives $\pi(n) \propto [zm(n)]^{1/(1-\gamma)}$.

It arises from $m(n) = \exp[-(1 - \gamma)\varphi(n - n_p)^2/2]$ in the production structure above; Online Appendix A.1 gives an illustrative foundation. The normalization makes φ the curvature of log profit after labor reoptimization. Its semi-elasticity is linear:

$$f(n) \equiv \frac{\pi'(n)}{\pi(n)} = -\varphi(n - n_p). \quad (3)$$

High φ means a narrow profit peak; low φ a broad one. This linearity delivers exact harmonic motion as a corollary of the general classification.

2.3 Reduced-form timing friction

The benchmark takes the timing friction as reduced form. It penalizes rapid *net industry reconfiguration*: expansion strains permitting and construction, while contraction strains resale, scrapping, and reorganization. The state derivative matters because it records how rapidly the inherited industry is being changed.

Assumption 3 (Timing friction).

The multiplicative timing friction takes the log-quadratic form

$$\xi(\dot{n}) = e^{-\frac{\mu}{2}\dot{n}^2}, \quad \mu > 0,$$

where μ is the friction parameter.

The functional form makes effective profit fall with the squared reconfiguration rate, with μ a free parameter. Any smooth symmetric friction has the local expansion $\ln \xi(\dot{n}) = -\frac{\mu}{2}\dot{n}^2 + O(\dot{n}^4)$. Symmetry is load-bearing for the exact closed form. Online Appendix A.2–A.3 gives a discrete-time motivation and a matching-based scale for μ ; neither is imposed as market clearing.

The net-flow specification is restrictive. With large simultaneous entry and exit, net reconfiguration understates traffic through a gross-entry pipeline. The model instead treats both expansion and contraction as costly changes in an inherited industry state. A separate-pipeline model is an extension, not an alternative interpretation of the present state variable.

2.4 Combined structure

Effective profit is

$$y_t = \pi(n_t) \exp\left(-\frac{\mu}{2}\dot{n}_t^2\right). \quad (4)$$

The general theorem uses only this expression and Assumption 2. Under (2), it becomes a bivariate log-quadratic kernel centered at $(n_p, 0)$.

2.5 The agent's problem

A searching agent chooses a date τ at which to enter. Upon entry, she operates a firm earning the effective profit $y_t = \pi(n_t) \xi(\dot{n}_t)$ of Section 2.4 per unit time, discounted at rate $r > 0$, with exogenous exit at rate $\delta > 0$. The value of entering at date τ is:

$$V(\tau) = \int_{\tau}^{\infty} e^{-(r+\delta)(t-\tau)} y_t dt. \quad (5)$$

$V(\tau)$ is the value of a single entry spell: operating from τ until stochastic exit. Re-entry after exit is a fresh participation decision. Under the ironing condition, the single-spell formulation is therefore without loss for the full-support paths studied here. The agent maximizes $V(\tau)$ over τ , taking the aggregate path $\{n_t\}$ as given. An agent who does not enter at date t can wait, with continuation value

$$V_t^{\text{search}} = \max \left\{ 0, \sup_{\tau \geq t} e^{-r(\tau-t)} [V(\tau) - \kappa] \right\}, \quad (6)$$

where κ is the per-firm real entry-resource cost from Assumption 1, paid at the moment of entry.

2.6 Equilibrium

An *entry policy* is a measurable function $e : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ specifying the flow rate of entrants at each date. Together with the initial condition n_0 , it determines the entire path of the economy through the aggregate consistency condition $\dot{n}_t = e_t - \delta n_t$.

Definition 1 (Full-support timing-indifference equilibrium).

A full-support timing-indifference equilibrium is a path $\{n_t\}_{t \geq 0}$ and an entry policy $e(\cdot)$ such that:

- (i) **Individual optimality:** Each participant maximizes discounted continuation value, comparing immediate entry against waiting and entering later, with waiting value V_t^{search} of (6).
- (ii) **Interior participation and timing indifference:** participation and entry are positive at every date, the outside reservoir is slack, and competitive participation gives $V_t^{\text{search}} = 0$ and $V(\tau) = \kappa$ for every τ .
- (iii) **Aggregate consistency:** $\dot{n}_t = e_t - \delta n_t$, where $e_t \geq 0$.
- (iv) **Feasibility:** $n_t < M$ for all t .

Interiority as a maintained restriction Ordinary participation gives complementary slackness: positive participation at a date implies $V_t = \kappa$, while zero participation permits

$V_t \leq \kappa$. Definition 1 studies the stronger case in which every date is active and the outside reservoir remains nonempty, so equality holds everywhere. The amplitude bound below verifies $e_t > 0$ and $n_t < M$ for the harmonic paths and stationary endpoints. It does not show that every equilibrium must have full support. Proposition 1 characterizes the consequence of this maintained interiority restriction.

Commitment versus dynamic optimization The formulation in (5)–(6) is a commitment problem: agents choose τ taking the future path $\{n_t\}$ as given. The ironing condition makes it equivalent to the dynamic problem in which agents observe $(n_t, \dot{n}_t)$ and decide whether to enter at each instant: since $V(\tau) = \kappa$ for all τ in equilibrium, an agent free to re-optimize finds no profitable deviation and chooses not to re-time.

2.7 Timing indifference

The next proposition turns the no-retiming idea into an equation. If the value of entry is the same at every date, then a small delay cannot change that value. The flow payoff must therefore exactly cover the annuity value of the entry cost. I call this condition *ironing*, because it flattens the payoff to entry across time.

Proposition 1 (Differential ironing on active dates).

Full-support timing indifference (Definition 1(ii)), $V(\tau) = \kappa$ for all τ , implies the differential ironing condition almost everywhere:

$$\pi(n_t) \xi(\dot{n}_t) = (r + \delta) \kappa \quad \text{for a.e. } t. \quad (7)$$

In a classical full-support equilibrium it holds at every date.

Proof. By Definition 1(ii), entry occurs at every date and $V(\tau) = \kappa$ for all τ . The bounded flow payoff makes $V(\tau)$ Lipschitz and hence differentiable almost everywhere.³ Differentiating (5):

$$\frac{dV}{d\tau} = -\pi(n(\tau)) \xi(\dot{n}(\tau)) + (r + \delta) V(\tau).$$

Since $V \equiv \kappa$, its derivative is zero and the displayed condition follows almost everywhere. In a classical full-support equilibrium, n and $\dot{n}$ are continuous, so it extends to every date. Because the symmetric friction treats $|\dot{n}|$ and $-|\dot{n}|$ identically, ironing determines speed, not the initial direction. ■

Proposition 1 is a deterministic, pathwise result for a constant entry-resource cost. A conditional slow-moving perturbation is reported separately in Online Appendix B.

³The first-order condition holds almost everywhere and extends to every date once the integrand has a continuous version.

3 Equilibrium analysis

3.1 The equilibria

The full-support definition is stronger than ordinary free-entry complementarity. At an inactive date the ordinary condition permits $V(t) \leq \kappa$ and $e_t = 0$; equality is required only when entry is positive. That distinction adds a collapse equilibrium, separate from any question of how a moving path continues.

On the active branch, timing indifference gives another source of multiplicity. Write the break-even flow payoff as $\bar{y} = (r + \delta)\kappa$ and define

$$h(n) \equiv \ln \frac{\pi(n)}{\bar{y}}.$$

Taking logs in (7) gives the constant-payoff locus

$$\frac{\mu}{2} \dot{n}_t^2 = h(n_t). \quad (8)$$

Assumption 2 implies that h is positive between n_- and n_+ , zero at those endpoints, and crosses zero with a nonzero slope. The locus confines the state to $[n_-, n_+]$ and fixes the absolute speed at every interior scale. Its endpoints are stationary full-support equilibria; the same locus also contains cyclical candidate paths.

Call a full-support timing-indifference equilibrium *classical* if $e \in C^1(\mathbb{R}_+)$, equivalently $n \in C^2(\mathbb{R}_+)$ because

$$e_t = \dot{n}_t + \delta n_t, \quad \dot{e}_t = \ddot{n}_t + \delta \dot{n}_t,$$

where e is the aggregate density of chosen entry dates. This regularity imposes smoothness on that aggregate density.

The locus is a strong restriction with a simple reading. Effective profit must be the same at every active date, but effective profit falls both with distance from the profit peak and with the speed of reconfiguration. An industry far from its peak must therefore be moving slowly, and an industry near its peak must be moving fast: what the industry gives up in scale it must make back in calm, and what it gains in scale it must pay for in haste. That is what pins speed to scale.

What the locus does not say is which way the industry travels along it, or whether it travels at all. The next subsection settles that in the log-quadratic benchmark, and the answer is three kinds of equilibrium rather than one.

Under the log-quadratic landscape (2), write

$$b \equiv \ln \frac{\pi_0}{\bar{y}} > 0, \quad A \equiv \sqrt{\frac{2b}{\varphi}},$$

so that the break-even scales are $n_- = n_p - A$ and $n_+ = n_p + A$ and the locus (8) becomes the ellipse

$$\frac{\varphi}{2} (n_t - n_p)^2 + \frac{\mu}{2} \dot{n}_t^2 = b. \quad (9)$$

The analysis focuses on equilibria with a smooth aggregate entry schedule: $e \in C^1(\mathbb{R}_+)$, equivalently $n \in C^2(\mathbb{R}_+)$ since $e_t = \dot{n}_t + \delta n_t$. Call these *classical*. The restriction rules out a kink in the density of chosen entry dates; Proposition 2 shows it is inessential for selecting the cycle.

Theorem 1 (Classification of equilibria).

Under Assumptions 1–3 and (2), the model admits three kinds of equilibria.

- (i) **Rest.** $n_t \equiv n_p - A$ and $n_t \equiv n_p + A$.
- (ii) **Cycle.** For any phase ϕ ,

$$n_t = n_p + A \cos(\omega t + \phi), \quad \omega = \sqrt{\frac{\varphi}{\mu}}, \quad T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{\mu}{\varphi}}, \quad (10)$$

provided

$$A < \min \left\{ \frac{\delta n_p}{\sqrt{\omega^2 + \delta^2}}, M - n_p \right\}. \quad (11)$$

- (iii) **Withering.** $e_t = 0$ and $n_t = n_0 e^{-\delta t}$, from any inherited n_0 , whenever $\bar{y} \geq \bar{y}_c \equiv \sup_{m \in [0, M)} \pi(m) \xi(-\delta m)$. Since $\bar{y}_c < \pi(n_p)$, the admissible interval $[\bar{y}_c, \pi(n_p))$ for the break-even level $\bar{y}$ is nonempty.

Proof. Let $U = \{t \geq 0 : \dot{n}_t \neq 0\}$. Differentiating (9) on U and cancelling $\dot{n}_t$ gives

$$\mu \ddot{n}_t + \varphi (n_t - n_p) = 0. \quad (12)$$

If U is empty, (9) leaves only $n_t \equiv n_p \pm A$. If U is nonempty, let t_0 be a boundary point of U : continuity and (9) give $\dot{n}_{t_0} = 0$ and $n_{t_0} = n_p \pm A$, and taking limits along U with $n \in C^2$ gives $\ddot{n}_{t_0} = \mp \varphi A / \mu \neq 0$. So t_0 is an isolated zero of $\dot{n}$, a relative neighbourhood of it lies in $\overline{U}$, and $\overline{U}$ is therefore both open and closed in the connected set $\mathbb{R}_+$. Hence U is dense and continuity extends (12) to every date. Its solutions are (10), with the amplitude fixed at A by (9) and the period $2\pi/\omega$.

For feasibility, $e_t = \delta n_p + A[\delta \cos(\omega t + \phi) - \omega \sin(\omega t + \phi)]$, whose minimum is positive exactly under the first bound in (11); the second keeps the outside reservoir slack. Each rest path has $e_t = \delta n_{\pm} > 0$. Along (i) and (ii) effective profit equals $\bar{y}$, so $V(\tau) = \kappa$ at every date.

For (iii), the flow payoff on the withering path is $\pi(n_t) \xi(-\delta n_t) \leq \bar{y}_c \leq \bar{y} = (r + \delta) \kappa$ at every date, so $V(\tau) = \int_{\tau}^{\infty} e^{-(r+\delta)(s-\tau)} \pi(n_s) \xi(-\delta n_s) ds \leq \kappa$ for every τ and complementary slackness makes $e_{\tau} = 0$ optimal throughout. That $\bar{y}_c < \pi(n_p)$ holds because the product equals $\pi(n_p)$ only if $\pi(m) = \pi(n_p)$ and $\xi(-\delta m) = 1$, i.e. $m = n_p$ and $m = 0$ together, which

$n_p > 0$ forbids: on the compact $[0, n_p]$ the continuous product attains a maximum strictly below $\pi(n_p)$, and on $[n_p, M)$ it is at most $\pi(n_p)\xi(-\delta n_p) < \pi(n_p)$. ■

Cases (i) and (ii) exhaust the classical full-support equilibria: every nonstationary one is (10), and the only stationary ones are the two rest paths. Case (iii) drops full support, since no date is active. Proposition 2 below selects case (ii) as the dynamically robust equilibrium, and the analysis focuses on it.

The three branches are three different answers to the same question, and it is worth seeing why each is coherent. On the cycle, entrants keep arriving through the expansion because profits are still above the entry cost, and they keep arriving past the peak because the alternative, stopping, would leave the industry too calm for the date-by-date indifference to hold. At a break-even scale the industry is already so far from its peak that profits exactly cover entry with no reconfiguration at all, so replacement entry alone sustains it. And an industry can wither for the reason that any coordination failure bites: entering an industry that nobody else enters means buying into a shrinking market, which is not worth the entry cost, and the expectation is self-confirming. An industry resting exactly at its profit peak, the most profitable configuration available to it, can die this way.

Intuition The frequency has a mechanical reading. Equation (12) is the equation of a mass on a spring, $\mu\ddot{n} = -\varphi(n - n_p)$: profit curvature φ is the *restoring force*, since a steeper profit peak punishes deviations harder and pulls the industry back faster, while the timing friction μ is the *inertia*, since a stiffer pipeline makes the industry slower to turn. A stiff spring with a light mass oscillates quickly; a slack spring with a heavy mass oscillates slowly. The industry does the same, at $\omega = \sqrt{\varphi/\mu}$, exactly as a mass-spring system with stiffness k and mass m runs at $\sqrt{k/m}$. Both primitives are needed: with no scale externality the spring has no stiffness, and with no friction it has no inertia.

The analogy also explains why the frequency is free of the orbit's size. Ironing fixes maximal speed at the profit peak and slower motion near the break-even scales, and in the log-quadratic case speed and distance scale together: a deeper orbit travels a longer way, but proportionally faster, and the two effects cancel. A displaced industry therefore takes the same time to come back whether it was displaced a little or a lot. The same cancellation makes the frequency free of productivity. A multiplicative increase in productivity lifts the whole profit curve, which enlarges the orbit at a fixed raw entry cost, but it leaves the *curvature* of log profit untouched (the spring is no stiffer for being higher off the ground), so the frequency does not move.

The full-support paths on the ellipse (9) are the harmonic cycle (10), the two rest paths $n_t \equiv n_{\pm}$, and dwelling paths that pause at a break-even scale before traversing. A tremble

in the entry flow at a date s displaces the phase-space state to some $(n_s^\varepsilon, \dot{n}_s^\varepsilon)$ within ε of $(n_s, \dot{n}_s)$; let $\{n_t^\varepsilon\}_{t \geq s}$ solve the continuation law (12) from it. A full-support equilibrium $\{n_t\}$ is *dynamically robust* if $\sup_{t \geq s} |n_t^\varepsilon - n_t| \rightarrow 0$ as $\varepsilon \rightarrow 0$, for every $s \geq 0$ and every such displacement.

Proposition 2 (Dynamic robustness selects the cycle).

The harmonic cycle (10) is the unique dynamically robust full-support equilibrium on (9).

Proof. A displacement at a break-even scale moves the state off $(n_\pm, 0)$ to a nearby level of (9), where $\ddot{n} = -\varphi(n - n_p)/\mu \neq 0$, so the continuation traces that ellipse: a swing of amplitude $2A + O(\varepsilon)$ that returns to the break-even scale only after a full period and with $\dot{n} \neq 0$ there. It therefore leaves every neighbourhood of the break-even scale and is $O(1)$ -far from any path that pauses there, rest or dwelling. A displacement on the cycle moves it to a level $b + O(\varepsilon)$, and its continuation is the harmonic path of that level, differing from the original in amplitude and phase by $O(\varepsilon)$. Because every level of (9) shares the angular frequency $\omega = \sqrt{\varphi/\mu}$, the two paths never drift out of phase, so the continuation stays uniformly $O(\varepsilon)$ -close over all t . ■

A rest path sits forever at a break-even scale, and a dwelling path pauses there before moving on; both rely on the industry being exactly still at a turning point, where the indifference condition holds only trivially. Nothing anchors it there. Let the entry flow tremble and the industry is in motion; the continuation law (12) then carries it across its whole range, a swing of amplitude $2A$, and it never again comes to rest. The cycle, already in motion, is only nudged to a neighbouring orbit; because every orbit shares the frequency ω , it neither changes period nor drifts out of phase, so the nudged path stays uniformly close. This global-in-time stability is special to the log-quadratic landscape: under a general profit curve a nudged orbit would run at a slightly different frequency and eventually drift out of phase, so isochrony is what makes the selected cycle stable as well as closed-form. Withering is different in kind, a coordination outcome in which no date is active, so no perturbation of an active industry reaches it. Timing indifference pins the speed at each scale; dynamic robustness picks the cycle out of the paths consistent with it, and the smooth-schedule restriction of Theorem 1 is inessential, since the dwelling paths it rules out are in any case not robust.

An adverse implication Amplitude-independent timing is a sharp restriction of the log-quadratic benchmark, and a direct panel diagnostic is adverse to it. Mapping recession depth to orbit amplitude, I regress log trough-to-recovery time on log peak-to-trough depth for 85 uncensored industry–recession episodes, with industry fixed effects and standard errors clustered by recession. The slope is 0.469 (s.e. 0.179). With seven recession clusters, the

normal-reference p -value is 0.009 and a t_6 reference gives $p = 0.040$. Forty-nine additional episodes are right-censored because production did not regain its prior peak within five years. Production is not the model’s establishment state, recession episodes need not trace classical equilibrium arcs, and censoring is nonrandom. The positive point estimate is therefore adverse evidence for exact isochrony, not a definitive test. The general period of the next paragraph allows amplitude dependence, so the diagnostic bears on the log-quadratic restriction and its empirical mapping rather than on every profit landscape. Online Appendix D reports the design.

Periodicity and isochrony The orbit is general; the constant frequency is not. The same argument runs on any single-peaked landscape of Assumption 2: the locus still confines the state to $[n_-, n_+]$, classical scheduling still forces reversal at the break-even scales, and every nonstationary classical full-support equilibrium is still a periodic orbit, now with period $T(\bar{y}) = \sqrt{2\mu} \int_{n_-}^{n_+} [\ln(\pi(n)/\bar{y})]^{-1/2} dn$. What the log-quadratic case adds is amplitude independence: the period stops depending on the orbit’s size, and the frequency collapses to $\sqrt{\varphi/\mu}$. That addition is less special than it looks. As $\bar{y} \uparrow \pi(n_p)$ the general period reproduces it locally, $2\pi/T(\bar{y}) \rightarrow \sqrt{-\pi''(n_p)/[\mu\pi(n_p)]}$, so every smooth peak is harmonic for small orbits. The log-quadratic landscape is the one that extends the small-orbit frequency to every amplitude. Online Appendix B.3 gives both statements.

3.2 Theoretical predictions

Conditional on the cyclical branch, the benchmark makes two predictions about frequencies. Neither prediction determines whether an industry belongs to the active or withering class. Conditional on full support, Proposition 2 selects the cyclical path; the empirical analysis asks whether its frequency implications appear in industries that fluctuate.

Prediction 1: industry specificity By (10) the frequency $\omega_i = \sqrt{\varphi_i/\mu_i}$ depends on the industry’s own primitives and on nothing else, so industries with different profit landscapes or entry pipelines cycle at different rates. The cross-industry spread of $\ln \omega_i$ should therefore exceed what estimation error alone would produce. This prediction requires no proxy and no auxiliary mapping: it is a statement about the estimated frequencies themselves.

Prediction 2: monotonicity Taking logs in (10), independent industries satisfy

$$\ln \omega_i = \frac{1}{2} \ln \varphi_i - \frac{1}{2} \ln \mu_i, \quad (13)$$

so a sharper log-profit peak raises frequency while a stiffer pipeline lowers it. The content is the *ordering*: holding the friction channel fixed, frequency should rise with profit curvature. Under general profits the period depends on the entire landscape at finite amplitude, and a skewed landscape makes the orbit spend unequal time on either side of n_p ; because the friction is symmetric in $\dot{n}$, the expansion and contraction legs remain exact time reversals of equal duration. Online Appendix B.3 and B.5 give the derivations; Online Appendix C reports numerical illustrations.

Taking Prediction 2 to data requires an additional mapping. Motivated by the geographic-concentration concept of Ellison and Glaeser (1997), I use a fixed, rounded industry-concentration ordering. The complete vector is archived, but its underlying observations, concordance, and aggregation weights are unavailable; it is neither a published nor an independently reproducible NAICS-3 Ellison–Glaeser series. I treat the score γ_i as a correlate of profit curvature, not as a direct measure. If $\ln \varphi_i = a + \lambda \ln \gamma_i + v_i$ and $\ln \mu_i$ is orthogonal to $\ln \gamma_i$, the population slope of $\ln \omega_i$ on $\ln \gamma_i$ is $\lambda/2$. The model therefore motivates a positive ranking under those restrictions; it does not imply that the implemented coefficient must equal one-half.

4 Empirical evidence

4.1 Industry specificity

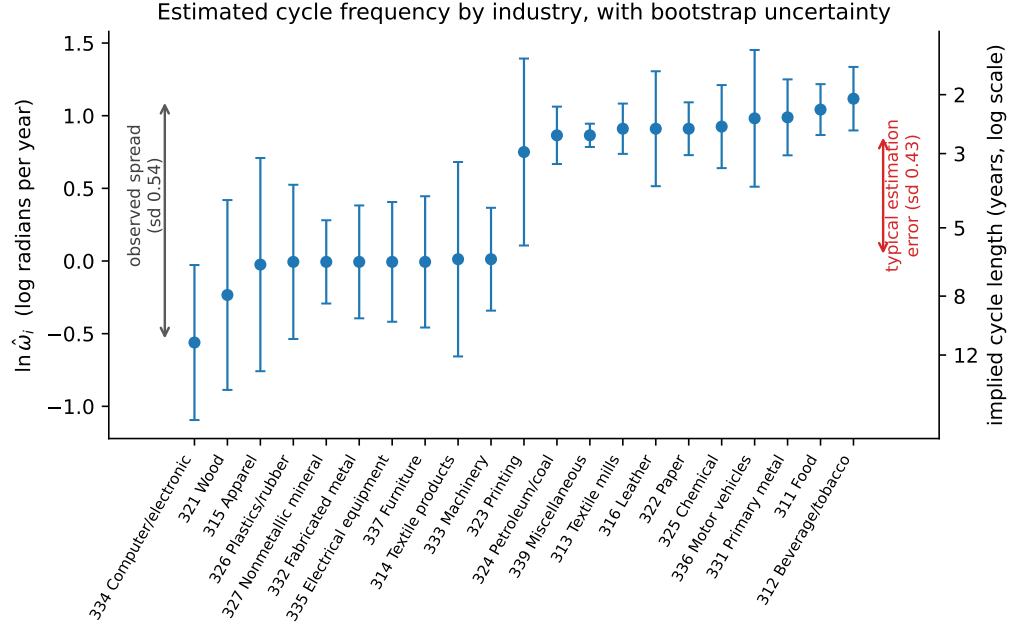
Prediction 1 concerns the estimated frequencies themselves, so it can be assessed before any proxy is introduced. For each of the 21 industries I estimate $\ln \hat{\omega}_i$ with the five-taper rule of Section 4.2, and I measure how precisely that peak is estimated by simulating 200 series from the industry’s own fitted AR(12) and re-estimating the peak on each.

Across the 21 industries the estimated cycle runs from 2.1 to 11.0 years (Figure 1), a wide spread under the common aggregate conditions the paper sets out to explain. The dispersion in log frequency, 0.54, exceeds the 0.46 estimation floor, as Prediction 1 requires. The margin over that floor is small, so the estimates place the industries in a broad pattern rather than a sharp ordering, but that pattern is the phenomenon the model is built to explain.

4.2 Cross-industry ranking

The headline design applies one pre-specified rule to all 21 NAICS-3 industries. For each I take the aggregate FRED production index over 1972–2024 (series IPGxxxS), estimate a five-taper Thomson DPSS spectrum of monthly log production growth, and select the largest peak in a fixed 2–12-year band. Concentration predicts frequency in the right direction: the slope of

Figure 1: Estimated cycle frequency by industry, with sampling uncertainty



Notes: Each point is $\ln \hat{\omega}_i$ from the five-taper rule on monthly log production growth, 1972–2024. Bars are ± 1 sampling standard deviation from the industry’s own parametric bootstrap. The arrows compare the spread of the estimates across industries with their typical sampling error.

log frequency on log concentration is 0.16 (HC1 s.e. 0.12), positive as Prediction 2 requires.⁴ Since production is not the model’s establishment state, Table 1 adds two establishment-based checks.

Table 1: Concentration and estimated industry frequency

	Production	Annual establishments	Quarterly establishments
Source	FRED IP	BDS	QCEW
Period	1972–2024	1978–2022	1990–2024
Industries	21	15	15
Peak estimator	DPSS, $K = 5$	Burg AR(6)	DPSS, $K = 5$
$\hat{\beta}_\gamma$	0.16	0.29	−0.024
HC1 s.e.	(0.12)	(0.11)	(0.013)

Notes: Each column regresses log estimated frequency on the log archived concentration score described in the text. Production and QCEW frequencies use the largest five-taper peak in the fixed 2–12-year band. BDS frequencies use log-differenced establishment counts and Burg AR(6) peaks in the same band; three of its 15 estimates lie at a band boundary. HC1 standard errors treat estimated peak locations as observed.

Annual BDS establishment log frequencies correlate 0.32 with production log frequencies under the current fixed-band protocol ($p = 0.24$) and produce the same positive concentration slope. Quarterly QCEW establishments do not: a common component accounts for a median

⁴The estimate is imprecise, not distinguishable from zero at conventional levels, and because the rule assigns a peak to every series it ranks protocol-specific frequencies rather than certifying a periodic component in each.

63% of establishment-growth variation, peak periods cluster between roughly 9.5 and 11.9 years, and the slope remains nonpositive using the historical instrument (-0.003) or after removing the common factor (-0.090). The annual counts reproduce the qualitative ranking, while the quarterly panel is a direct-state non-confirmation.

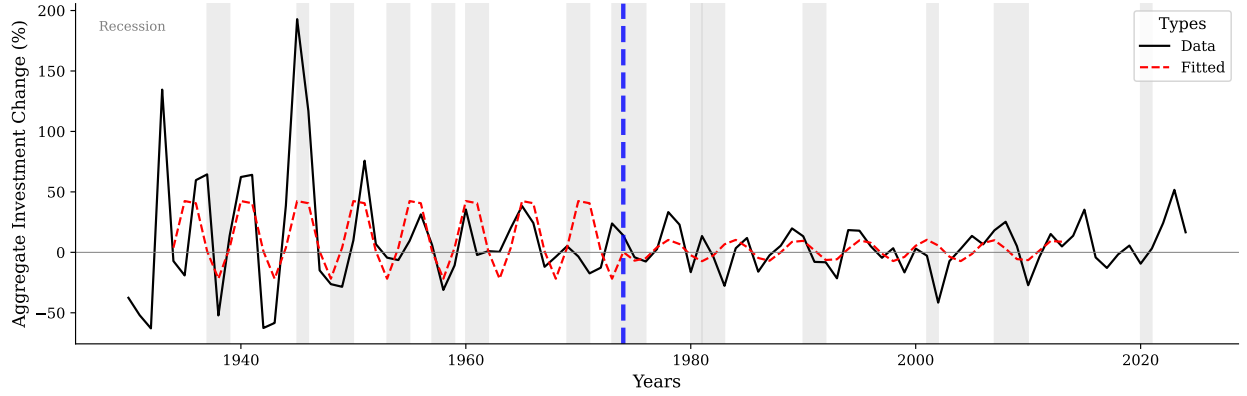
Screened 15-industry designs give larger but peak-selection-sensitive estimates; that estimator panel is in the Online Appendix. Together these exercises support a model-inspired ranking without identifying φ_i , μ_i , or the concentration-to-curvature mapping.

4.3 Historical echoes

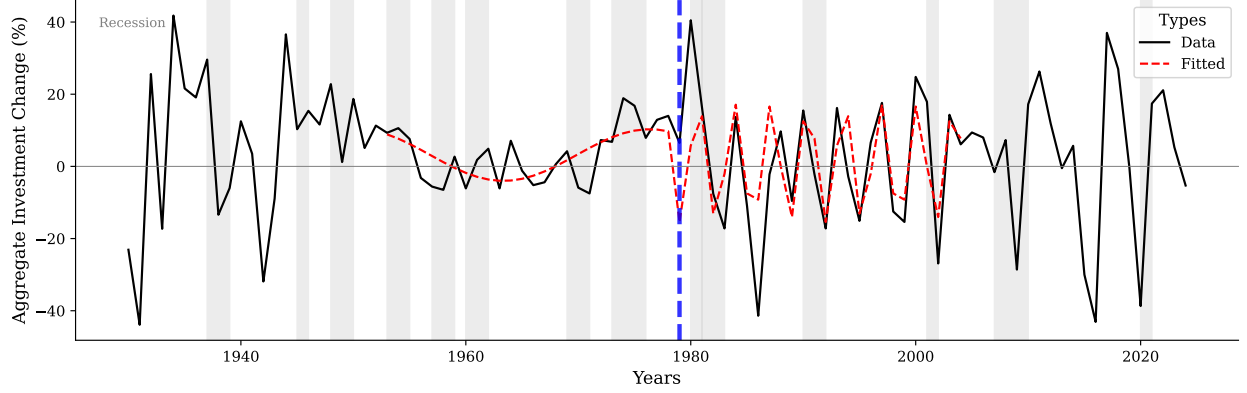
Two long historical episodes show the regularity the theory is about. Across decades of otherwise irregular history, manufacturing and oil investment alternate at steady, industry-specific rhythms. The industries, windows, and specification were chosen after inspecting the episodes, so these are illustrations rather than tests; their value is to make differing rhythms visible.

Manufacturing investment in 1934–1973 carries a five-year component, and mining investment after the 1979 oil crisis a 3.25-year one (Figure 2). These are investment flows rather than the model’s establishment stock. Online Appendix D reports the construction of the two series and the window definitions.

Figure 2: Fitted periodic components in two historical investment windows



(a) Manufacturing structures



(b) Mining exploration, shafts, and wells

Notes: Annual growth in private fixed investment from BEA NIPA Table 5.4.1. Solid line: data; red dashed line: fitted cosine; gray bands: NBER recessions. Blue dashed lines separate each event window from its equal-length comparison window. Within each window, the frequency maximizes the demeaned Fourier periodogram and the cosine is fitted by OLS. The event-window periods are $T = 5.0$ years for manufacturing in 1934–1973 and $T = 3.25$ years for mining/oil in 1979–2004. These components do not distinguish timing search from vintage replacement, structural breaks, or other propagation mechanisms.

5 Pigouvian taxation

The normative benchmark is the planner's resource problem. Under the paper's linear-surplus accounting, the planner chooses gross entry to maximize

$$\int_0^\infty e^{-\rho t} [n_t \pi(n_t) \xi(\dot{n}_t) - \kappa e_t] dt$$

subject to $\dot{n}_t = e_t - \delta n_t$. Locally, a rest optimum n^* satisfies

$$\pi(n^*) + n^* \pi'(n^*) = (\rho + \delta) \kappa. \quad (14)$$

Free entry rests where an entrant's own profit just covers the cost, $\pi(n) = (\rho + \delta) \kappa$; the planner's condition carries the extra term $n^* \pi'(n^*)$, the profit a marginal entrant adds to,

or removes from, incumbents through the scale channel. Free entry ignores this externality. Above the profit peak the term is negative, an extra firm congesting the market, so free entry brings in too many firms and the planner would tax entry; below the peak the sign reverses. Proposition 3 prices the externality state by state, along the whole path rather than only at rest, so that private free entry reproduces the planner's allocation.⁵

Proposition 3 (State-dependent Pigouvian wedge).

Along an interior planner path, the state-dependent rebated entry wedge

$$\tau_t = -n_t \pi(n_t) \xi'(\dot{n}_t) + \int_t^\infty e^{-(\rho+\delta)(s-t)} [\delta n_s \pi(n_s) \xi'(\dot{n}_s) - n_s \pi'(n_s) \xi(\dot{n}_s)] ds \quad (15)$$

decentralizes that interior planner allocation. At a rest point,

$$\tau^* = -\frac{n^* \pi'(n^*)}{\rho + \delta}.$$

It is positive for a congestion-side optimum with $n^ > n_p$.*

Proof. Take entry as the control, with current-value Hamiltonian

$$\mathcal{H} = n \pi(n) \xi(e - \delta n) - \kappa e + \lambda (e - \delta n), \quad \dot{n} = e - \delta n.$$

The entry margin $\partial \mathcal{H} / \partial e = 0$ gives

$$\lambda_t = \kappa - n_t \pi(n_t) \xi'(\dot{n}_t), \quad (16)$$

and $\dot{\lambda} = \rho \lambda - \partial \mathcal{H} / \partial n$ gives the costate equation

$$\dot{\lambda}_t = (\rho + \delta) \lambda_t - [\pi(n_t) + n_t \pi'(n_t)] \xi(\dot{n}_t) + \delta n_t \pi(n_t) \xi'(\dot{n}_t),$$

which, under $e^{-(\rho+\delta)T} \lambda_T \rightarrow 0$, integrates forward to

$$\lambda_t = \int_t^\infty e^{-(\rho+\delta)(s-t)} \left\{ [\pi(n_s) + n_s \pi'(n_s)] \xi(\dot{n}_s) - \delta n_s \pi(n_s) \xi'(\dot{n}_s) \right\} ds. \quad (17)$$

By (16) the first term of (15) is $\lambda_t - \kappa$, so substituting (17) collects the two integrals into one. Their integrands sum to

$$\{(\pi + n\pi')\xi - \delta n\pi\xi'\} + \{\delta n\pi\xi' - n\pi'\xi\} = \pi(n_s) \xi(\dot{n}_s),$$

the scale and replacement terms cancelling, so that $\tau_t = \int_t^\infty e^{-(\rho+\delta)(s-t)} \pi(n_s) \xi(\dot{n}_s) ds - \kappa = V_t - \kappa$, using $r = \rho$. Hence $V_t = \kappa + \tau_t$: the planner's path satisfies private free entry under (15). At a rest point $\xi'(0) = 0$ removes the first term and the integral reduces to $-n^* \pi'(n^*) / (\rho + \delta)$. ■

The wedge splits into the two channels through which an entrant affects others. The first term, $-n_t \pi(n_t) \xi'(\dot{n}_t)$, prices the reconfiguration the entrant adds at the moment of entry:

⁵The planner's optimum is a rest point to leading order, which requires concavity of $n\pi(n)\xi(\dot{n})$ rather than of π alone. Online Appendix E gives the derivation and the flat-tax implementation.

since $\xi' = -\mu\dot{n}\xi$, it taxes entry while the industry expands and subsidizes it while it contracts. The integral capitalizes the scale externality $-n\pi'$ imposed on incumbents over the entrant's life, net of its interaction with replacement entry. At a rest point the first term vanishes, leaving the static Pigouvian tax $\tau^* = -n^*\pi'(n^*)/(\rho + \delta)$.

Multiplicity under the flat tax The separate cycle-eliminating tax

$$\tau^{\text{collapse}} = \frac{\pi(n_p)}{r + \delta} - \kappa$$

raises the normalized private entry wedge to one. Under full-support ironing, the payoff locus then shrinks to the stationary point $(n_p, 0)$ because $\pi(n) \leq \pi(n_p)$ and $\xi(\dot{n}) \leq 1$, with joint equality only there. Under ordinary free-entry complementarity, however, the same tax also supports extinction from any inherited n_0 : set $e_t = 0$ and $n_t = n_0 e^{-\delta t}$. Because $\pi(n_t)\xi(-\delta n_t) \leq \pi(n_p)$, entry value never exceeds the taxed cost $\kappa + \tau^{\text{collapse}} = \pi(n_p)/(r + \delta)$. Thus the same flat tax supports both parking at n_p and extinction from any inherited scale. It generally does not implement n^* and is not the state-dependent Pigouvian wedge in Proposition 3.

6 Concluding remarks

This paper studies equilibrium when entrants search over dates and every active date must offer the same value. The model admits collapse and active rest and, when the full-support feasibility bound holds, cycling. Within the classical full-support cyclical class, every nonstationary equilibrium is the same orbit up to phase and therefore has the same period.

The general profit curve fixes that period through an exact integral, and log-quadratic profits collapse it to the closed form $\omega = \sqrt{\varphi/\mu}$, a transparent profit-heterogeneity channel across industries. Taken to U.S. manufacturing, the concentration ranking is positive in the predicted direction and annual establishment frequencies share its sign; the quarterly panel does not, and the depth diagnostic is adverse to exact isochrony. The historical echoes make the frequency object visible.

Three boundaries matter for interpretation. First, the symmetric friction concerns net reconfiguration of an inherited industry. It misses heavy simultaneous gross entry and exit, so a separate gross-entry and exit structure need not share the same frequency formula. Second, the archived concentration score is not measured profit curvature and may covary with the timing friction. Third, the historical echoes are selected illustrations rather than inference.

The clean empirical next step is to measure both primitives independently: profit curvature from profits, markups, or survival as industry scale changes, and timing friction

from permitting, construction, equipment-delivery, or financing delays. On the theory side, endogenous exit and separate gross-entry and exit pipelines would clarify the range over which net reconfiguration is a sufficient state. Nonlinear household welfare and production networks are subsequent extensions.

Equal private values across dates do not price the scale and reconfiguration effects on incumbents, which yields the state-dependent Pigouvian wedge. But the simpler flat tax likewise supports both parking at n_p and extinction. Timing search accounts for industry frequency with a single closed-form expression in two primitives, $\omega = \sqrt{\varphi/\mu}$.

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